Banking Loan Risk Analysis Application

Technical Evaluation, Test Benchmark & Insights Report

1. Evaluation Methodology

The system was evaluated across data pipeline robustness, reconciliation accuracy, risk engine classification precision, API response latency, and end-to-end integration across the stack.

2. Automated Test Suite Results

A 14-test automated test suite was executed via Pytest covering all key modules:

- **ETL Pipeline Tests:** Verified raw extraction, categorical decoding, upsert semantics, and table persistence (100% Passed).
- **Business Rules & Risk Engine:** Verified Low, Medium, and High risk band classifications across edge cases, as well as high-exposure and chronic delinquency review triggers (100% Passed).
- **Data Quality & Reconciliation:** Validated duplicate checks, range checks, MD5 checksum computation, and record reconciliation audits (100% Passed).
- **SQL & Relational DB Tests:** Validated ACID CRUD operations, foreign key cascades, and pre-aggregated KPI tables (100% Passed).
- **FastAPI Endpoints:** Validated JWT RBAC, paginated querying, risk evaluation, and KPI aggregations (100% Passed).

3. Pipeline Performance Metrics

- **Sample ETL (200 records):** Execution time: 0.21s; Reconciliation: Reconciled (100%).
- **Incremental ETL (3,000 records):** Execution time: 3.19s; Reconciliation: Reconciled (100%).
- **Data Quality Checks:** 8 distinct rule suites executed per batch with 0 unhandled exceptions.
- **API Latency:** Under 25ms for paginated customer queries; under 15ms for risk evaluation.

4. Key Analytical Insights

- **Education Breakdown:** High School segment exhibited the highest default rate (26.17%) and highest high-risk concentration (48.88%), compared to Graduate School clients (19.23% default rate).
- **Age Trends:** Default rate peaks in the 60+ age group (28.57%) and 50-59 group (26.96%), while lowest in the 30-39 bracket (20.96%).
- **Utilization Indicator:** Customers with credit utilization above 85% accompanied by payment delays exhibited default rates exceeding 68%, confirming the validity of the High Risk classification rule.